Prognostic Value of Frailty in Aortic Surgery: A Systematic Review and Meta-Analysis Comparing Frailty Assessment Tools

Mayara Leite Coutinho,[1] Lucas Monteiro Delgado,[2] Inez Ohashi Torres,[1] João Rocha-Neves,[3] Antonio Eduardo Zerati,[1] Mario D'Oria,[4] and Pedro Puech Leão,[1] São Paulo, and Minas Gerais, Brazil, Portugal, and Trieste, Italy

Background: Frailty is increasingly recognized as an important determinant of outcomes after aortic vascular surgery, but assessment methods vary substantially and the optimal tool for risk stratification remains uncertain. This systematic review and meta-analysis evaluated the prognostic value of preoperative frailty and compared the predictive performance of different frailty instruments in aortic surgery.

Methods: PubMed, Embase, and Cochrane Library were searched from inception to April 27, 2026. Eligible studies included patients undergoing open, endovascular, or hybrid aortic procedures involving abdominal, thoracic, thoracoabdominal, arch, and proximal aortic diseases, including aneurysms and dissections, assessed frailty preoperatively, and reported postoperative outcomes.

Results: Thirty studies comprising 419,459 patients were included. Frailty was associated with higher early mortality (odds ratio [OR] 2.20; 95% confidence interval [CI] 1.54−3.14) and late mortality (hazard ratio 2.18; 95% CI 1.64−2.90). Frail patients also had increased risks of major complications (OR 2.52; 95% CI 1.22−5.19), acute kidney injury (OR 1.64; 95% CI 1.34−2.02), and nonhome discharge (OR 5.50; 95% CI 3.05−9.92). Associations were consistent across surgical approaches and aortic segments. Judgment-based or phenotype-like tools yielded higher effect estimates than deficit-accumulation indices, although differences were not statistically significant; among index-based tools, Modified Frailty Index (mFI)-11 outperformed mFI-5.

Conclusion: Preoperative frailty strongly predicts mortality, morbidity, and loss of functional independence after open, endovascular, and hybrid aortic surgery across different aortic segments and pathologies, including aneurysmal and dissecting aortic disease. Routine frailty assessment may improve risk stratification and perioperative decision-making.

Conflict of interest: The authors declare no conflicts of interest. No datasets were generated during the conduct of this review.

Funding: This study received no financial or nonfinancial support.

Availability of data: Additional data and materials related to the analysis are available upon reasonable request.

Registration: PROSPERO (CRD420261386881).

[1]*Department of Vascular Surgery, Hospital das Clínicas FMUSP, Faculdade de Medicina da Universidade de São Paulo, São Paulo, Brazil.*

[2]*Department of Medicine, Universidade Federal de Minas Gerais, Belo Horizonte, Minas Gerais, Brazil.*

[3]*Faculty of Medicine of the University of Porto, Unity of Anatomy, Department of Biomedicine, Portugal.*

[4]*Division of Vascular and Endovascular Surgery, Department of Clinical Surgical and Health Sciences, University of Trieste, Trieste, Italy.*

Correspondence to: Mayara Leite Coutinho, MD, Department of Vascular Surgery, Hospital das Clínicas FMUSP, Faculdade de Medicina da Universidade de São Paulo, Street, 255 - Dr. Enéas Carvalho de Aguiar, São Paulo, São Paulo 05405-000, Brazil; E-mail: mayara.leite84@hotmail.com

Ann Vasc Surg 2026; 132: 200–213

https://doi.org/10.1016/j.avsg.2026.06.034

© 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Manuscript received: May 25, 2026; manuscript accepted: June 19, 2026; published online: 29 June 2026

INTRODUCTION

Aortic vascular surgery encompasses a broad spectrum of open, endovascular, and hybrid interventions for abdominal aortic aneurysms (AAAs), descending thoracic aortic aneurysms (DTAAs) and thoracoabdominal aortic aneurysms (TAAAs), type B aortic dissection, and other complex aortic pathologies. These procedures impose substantial physiological stress and are frequently performed in older patients with multiple comorbidities and limited physiological reserve.[1,2] Indeed, as the population ages, an increasing proportion of patients presenting for aortic surgery are elderly and living with frailty,a multidimensional syndrome characterized by decreased physiological reserve and diminished resistance to stressors.[3,4] The prevalence of frailty among older surgical patients ranges from 25% to 40%, substantially higher than the 10.7% prevalence observed in community-dwelling adults aged $\geq$ 65 years.[3,4]

Accumulating evidence demonstrates that frailty is a powerful independent predictor of adverse outcomes following vascular surgery. Frailty has been associated with 30-day mortality, postoperative complications, and long-term mortality.[1,2,5,6] Specifically in AAA repair, frailty increases the risk of 30-day mortality approximately 4-fold and doubles the risk of long-term mortality.[5,6] Beyond traditional clinical outcomes, frailty predicts patient-centered outcomes including nonhome discharge, hospital readmission, and reduced health-related quality of life.[6,7]

Despite the established prognostic value of frailty, a major challenge in synthesizing the evidence is the substantial heterogeneity in assessment methods. At least 35 different frailty instruments have been used in perioperative research, broadly categorized into phenotypic approaches (e.g., Fried phenotype), deficit accumulation models (e.g., Frailty Index [FI]), and single-domain measures (e.g., functional status, central muscle mass).[1,2,8,9] These tools vary considerably in their components, complexity, feasibility, and predictive performance.

The current guidelines recommend preoperative frailty assessment using validated tools in patients of $\geq$ 65 years undergoing elevated risk surgery.[4,10,11] However, the optimal frailty assessment tool for aortic surgery remains unclear. Previous systematic reviews have examined frailty in broader vascular surgery or cardiac surgery populations, but comprehensive synthesis specifically comparing the predictive performance of different frailty assessment tools in aortic surgery is lacking.[1,2,7,12,13] Understanding which tools best predict outcomes in this high-risk population could inform preoperative risk stratification, guide shared decision-making, and identify patients who might benefit from prehabilitation or enhanced perioperative care.[4,9,14]

Therefore, this systematic review and meta-analysis aimed to evaluate the association between preoperative frailty and postoperative outcomes in aortic surgery patients. This meta-analysis seeks to provide robust evidence to guide clinical decision-making and improve outcomes in this high-risk population.

MATERIALS AND METHODS

Protocol and Registration

We performed the systematic review and meta-analysis according to the Cochrane Handbook for Systematic Reviews of Interventions and structured it according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.[15,16] The study protocol was registered in the International Prospective Register of Systematic Reviews under registration number CRD4202613868817.[17]

Eligibility Criteria

Studies were included if they met any of the following criteria: (1) evaluated patients undergoing aortic vascular surgery (open, endovascular, or hybrid), including AAA, DTAA/TAAA and type B aortic dissection; (2) preoperative frailty was assessed using validated instruments, such as the Clinical Frailty Scale (CFS), Frailty Index (FI), Modified Frailty Index (mFI), or other recognized measures; (3) separated outcomes by frailty and nonfrailty status; and (4) reported at least one of the outcomes of interest.

Studies were excluded if they met any of the following criteria: (1) did not specify which vascular segment was involved; (2) overlapping populations; (3) unavailable full text; or (4) publication type such as case reports, conference abstracts, meta-analyses, review articles, or animal studies.

Search Strategy and Study Selection

We systematically searched PubMed, Embase, and Cochrane Library databases from inception to April 27, 2026. The search strategy for all databases was: (frailty OR frail OR ''frailty index'' OR ''clinical frailty scale'' OR mFI) AND (''vascular surgery'' OR endovascular OR EVAR OR TEVAR OR ''aortic aneurysm'') AND (mortality OR complications OR outcomes OR prognosis OR survival). We also searched the references of the included studies and

previous systematic reviews and meta-analyses aiming for the inclusion of additional studies.[18]

Two authors (M.L.C and L.M.D.) independently conducted the search, imported results into Rayyan, a web-based systematic review tool, and triaged the studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the review of the full-text articles. Disagreements were solved by consensus.

Outcomes and Subgroup Analysis

The outcomes of interest were early mortality, defined as prior to 90-day or in-hospital mortality; late mortality, defined as mortality occurring after 90 days; major postoperative complications; nonhome discharge; and acute kidney injury (AKI). We performed subgroup analyses to explore potential sources of heterogeneity. These analyses were stratified according to the frailty assessment tool (FI-type versus judgment-based/phenotype-like), affected aortic segment (TAAA versus AAA versus thoracic only), and surgery type (endovascular versus open). Additionally, we further specified the FI-type scale (mFI-5 versus mFI-11), and for the AAA group specifically, we also assessed the type of frailty assessment.

Data Extraction

Two authors (M.L.C. and I.O.T.) independently extracted data from the included studies using a standardized data extraction form. The following information was collected: (1) general study characteristics, including first author, year of publication, country, study design, and study period; (2) patient and clinical characteristics, including total number of patients, mean or median age, and sex distribution; (3) procedural and disease-related variables, including type of aortic pathology, type of procedure, and clinical setting; (4) frailty-related variables, including frailty assessment tool, frailty cutoff definition, proportion of frail patients, and duration of follow-up; and (5) reported outcomes.

Statistical Analysis

Treatment effects for binary outcomes were pooled using odds ratios (ORs) for early mortality and hazard ratios(HR) for late mortality, both with corresponding 95% confidence intervals (CIs). Between-study heterogeneity was assessed using Cochran's Q test and the I^2 statistic. P values < 0.10 and I^2 values of 25%, 50%, and 75% were interpreted as indicating low, moderate, and high heterogeneity, respectively. Random-effects models were fitted using the restricted maximum likelihood estimator. CI for pooled estimates was calculated using the Hartung-Knapp-Sidik-Jonkman method. For subgroup analysis, CI for pooled estimates as calculated using the Wald method, and $P > 0.1$ was deemed significant. All statistical analyses were performed using R statistical software version 4.4.3 (R Foundation for Statistical Computing, Vienna, Austria), using the *meta* package.

Sensitivity Analysis

For outcomes with substantial heterogeneity ($I^2 \geq 25\%$), we performed additional sensitivity analyses using 4 distinct strategies to assess the robustness of our findings and explore potential sources of heterogeneity. First, leave-one-out analysis was performed, whereby each study was sequentially excluded to determine whether the pooled estimate was dependent on any single study. This analysis provides insight into the stability of the overall estimates and whether individual studies exert undue influence on the results.[19] Second, we conducted sensitivity analyses excluding studies judged to be at high risk of bias, to evaluate whether methodological limitations influenced the direction or magnitude of the pooled effects.

Risk of Bias Assessment

The risk of bias of each included study was assessed with the Risk of Bias in Nonrandomized Studies of Interventions tool.[20] Two independent authors conducted the risk of bias assessment (M.L.C. and L.M.D.), and disagreements were solved by consensus. The certainty of evidence was rated using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach and categorized as high, moderate, low, or very low.[21] Publication bias was assessed when at least 10 studies reported the same outcome, using visual inspection of contour-enhanced funnel plots and formal testing with Egger's regression asymmetry tests.[15,22]

RESULTS

Study Selection

As detailed in Figure 1, the initial search identified 3,728 records. After removal of 908 duplicate records and screening of titles and abstracts, 128 studies were selected for full-text review according to prespecified eligibility criteria. Of these, 98

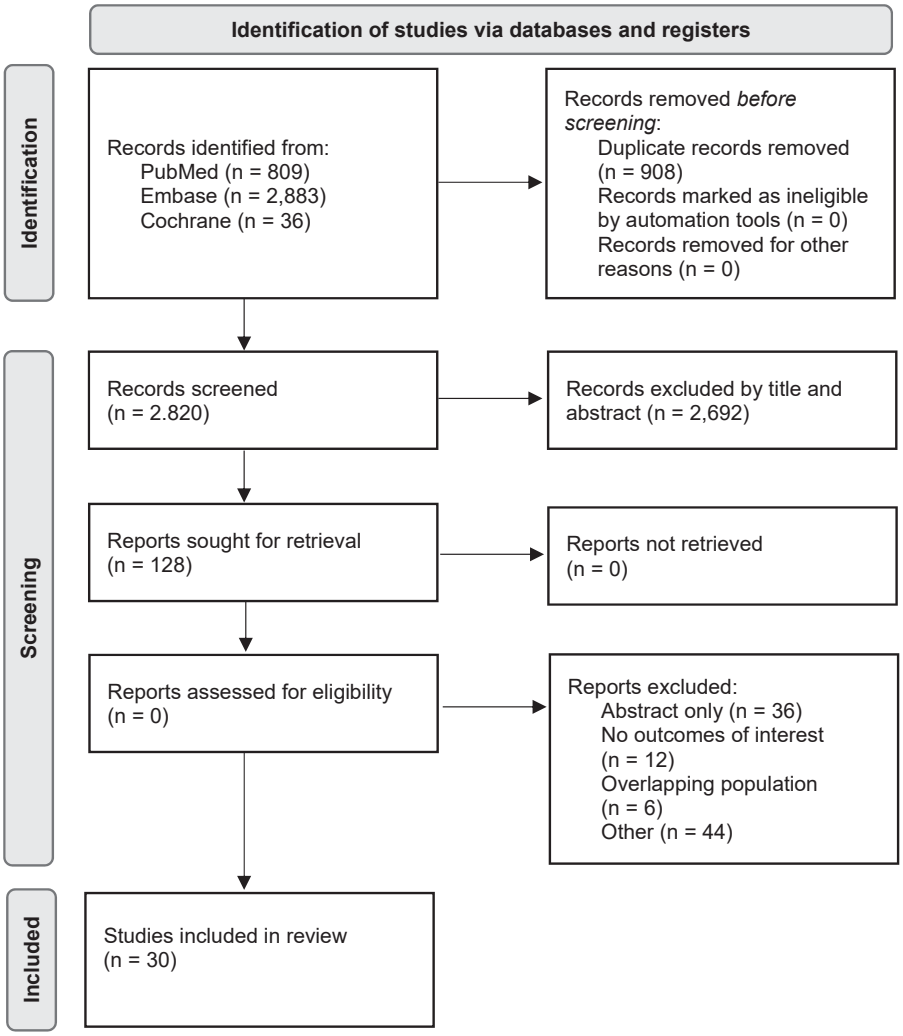


Fig. 1. PRISMA flow diagram. Study identification, screening, eligibility assessment, and inclusion process. PRISMA, preferred reporting items for systematic reviews and meta-analysis.

studies were excluded for the following reasons: abstract only (n = 36), no outcomes of interest (n = 12), overlapping population (n = 6), and other reasons (n = 44). Supplementary Table I lists the excluded studies from the full-text screening stage and the reasons for their exclusion.

Study and Patient Characteristics

A total of 30 studies, spanning from 2014 to 2025, were included.[23–52] Four of them were prospective, and the remaining were retrospective. A total of 419,459 patients were included in the analysis. Most of the patients were males (308,103, 69.9%), with a mean age ranging from 62.3 to 80 years. AAA was the most common disease, and most of the procedures were elective. Fourteen studies reported frailty assessment tools based on deficit accumulation (FI-type) and eight reported tools based on clinical/global frailty (judgment-based/phenotype-like). The most predominant frailty assessment tool in the FI-type group is mFI-5, and in the judgment-based/phenotype-like group, it is CFS. Study characteristics and patient demographics are detailed in Table I.

Pooled Analyses

Frailty was associated with significantly higher odds of both short-term and long-term mortality following aortic surgery. The pooled analysis for early mortality yielded an OR of 2.20 (95% CI 1.54−3.14; P = 0.0002; I^2 = 92.2%; Fig. 2), while the odds for late mortality remained high with an

Table I. Characteristics of included studies

Study	Country	Period	Design	No. of patients	Affected segment	Procedure	Setting	Mean age, years	Male, %	Frailty assessment tool	Frailty cutoff	Frail, %	Follow-up, months
Balasundaram 2023	USA	2015–2019	Retrospective	10,450	AAA	EVAR	Elective	73.6	80.2	mFI-5	≥0.6	4.3	1
Barbey 2019	USA and Canada	2003–2017	Retrospective	20,750	AAA	EVAR and OAR	Elective	73.3 (EVAR) 69.2 (OAR)	81.7 (EVAR) 74.4 (OAR)	mFI	0–9	NA	12
Chatterjee 2021	USA	2006–2016	Retrospective	392	TAAA	OAR	Mixed	69	60.5	Sarcopenia	Lower tertile	33.4	30
Chen 2023	USA	2000–2020	Retrospective	314	AAA	EVAR and OAR	Mixed	72.1 (EVAR) 70.2 (OAR)	99.5 (EVAR) 100 (OAR)	mFI-5	≥2	40.1	180
Chen 2025	USA	2012–2024	Retrospective	233	TAAA	F/BEVAR	Elective	72.8	72.5	CFS	≥4	25.7	60
Ciamarella 2021	USA	2011–2019	Retrospective	60	AAA	EVAR and OAR	Urgence	74	70	mFI-11	≥0.27	35%	1
Ganapathi 2014	USA	2005–2012	Retrospective	574	TAA	OAR	Mixed	65	59.6	Composite frailty	≥2	25.78	12
George 2020	USA	2003–2017	Retrospective	15,803	AAA	EVAR and OAR	Elective	68.9 (EVAR) 73 (OAR)	73.9 (EVAR) 80.7 (OAR)	VQI-RAI	≥30	12.4	48
Gomibuchi 2018	Japan	2004–2017	Retrospective	310	ATAAD	OAR	Urgence	67.5	50	Composite frailty	≥3	34.19	60
Kim 2021	USA	2005–2018	Retrospective	301,869	AAA + TAA	EVAR + TEVAR	Elective	73	79.9	ACG	Any frailty	2.5	Inhospital
Hiraoka 2018	Japan	2012–2017	Retrospective	113	Aortic arch disease	OAR	Elective	72.6	75.2	Frailty-based score	NA	NA	36
Kishimoto 2022	Japan	2010–2019	Retrospective	84	TAA	EVAR and OAR	Elective	80	78.6	CSHA	≥4	47.6	47
Ma 2025	USA	2005–2019	Retrospective	50,798	AAA	EVAR	Elective	73.3	80.6	mFI-11	≥0.27	NA	1
Mascia 2023	Italy	2010–2021	Retrospective	221	AAA	Open repair	Elective	72.2	89.1	mFI-5	>0.25	45	53
McGevna 2024	USA	2011–2022	Retrospective	78	AAA	EVAR and OAR	Urgence	75	70.5	mFI-11	≥0.27	44.9	7
Modrall 2022	USA	2011–2017	Retrospective	2,454	TAAA	EVAR and OAR	Elective	75	71	mFI-5	Continuous	NA	1

Murphy 2025	USA	2012–2023	Retrospective	213	TAAA	F/BEVAR	Mixed	73.3	72.3	CFS	≥4	26.7	60
Nobrega 2022	Portugal	2013–2020	Prospective	109	Aortoiliac occlusive disease	EVAR and OAR	Elective	62.6	94.5	mFI-5	>2	33	60
Paajanen 2022	Finland and USA	2007–2019	Retrospective	592	Complex AAA/ TAAA	EVAR	Elective	75	74	mFI-11	>0.5	14	60
Partridge 2015	UK	2011	Prospective	125	Mixed vascular	EVAR and OAR	Mixed	76	68.8	EFS	≥6.5	52	Inhospital
Percy 2024	USA	2016–2017	Retrospective	5,745	TAA	OAR	Mixed	62.3	68.6	ACG	Any frailty	7	Inhospital
Shakarchi 2020	UK	2014–2018	Retrospective	184	AAA	OAR	Elective	71.9	90.2	CFS	CFS = 4 vulnerable	16	1
Tak 2025	Turkey	2022–2024	Retrospective	155	Mixed vascular	EVAR and OAR	Elective	71.98	87.1	mFI-11	Continuous/ ≥3	NA	1
Thillainadesan 2020	Australia	2019	Prospective	150	Mixed vascular	EVAR and OAR	Mixed	79.5	68	FI/CFS	FI > 0.25/ CFS≥5	22.6	1
Tse 2022	USA	2001–2018	Retrospective	5,568	AAA	EVAR	Elective	71	99.6	RAI	Continuous cohorts	NA	1
Tsushima 2024	Japan	2011–2021	Retrospective	175	TAA	TEVAR	Elective	71.8	81.6	CFS	≥5	20.9	60
Visser 2019	Netherlands	2010–2017	Prospective	749	Mixed vascular	EVAR and OAR	Elective	72.1	77.6	GFI	≥4	29.2	1
Wang 2021	China	2011–2019	Retrospective	564	AAA	EVAR	NA	73	83.6	mFI	>2	23.7	32
Wang 2025	China	2016–2024	Retrospective	174	AAA	EVAR	Elective	72	88.5	mFI-5	≥2	32.7	26
Zou 2024	China	2015–2020	Retrospective	453	AAA	EVAR	NA	73	85.2	GNAI	Tertiles/ < 99	38.6	28

AAA, abdominal aortic aneurysm; EVAR: endovascular aneurysm repair; OAR: open aortic repair; TAAA: thoracoabdominal aortic aneurysm; F/BEVAR, fenestrated/branched endovascular aneurysm repair; TEVAR, thoracic endovascular aortic repair; mFI, Modified Frailty Index; CFS, Clinical Frailty Scale; RAI, Risk Assessment Index; VQI-RAI, Vascular Quality Initiative-Risk Assessment Index; ACG, Johns Hopkins Adjusted Clinical Groups; CSHA, Canadian Study of Health and Aging; FI, Frailty Index; GFI, Geriatric Frailty Index; GNAI, Geriatric Needs Assessment Index; mFI-5, Modified Frailty Index with five components; mFI-11, Modified Frailty Index with 11 components; mFI-5 (≥0.6), Modified Frailty Index with a cutoff of ≥ 0.6; mFI-5 (> 0.25), Modified Frailty Index with a cutoff of > 0.25; EFS, Edmonton Frail Scale; ATAAD, acute type A aortic dissection.

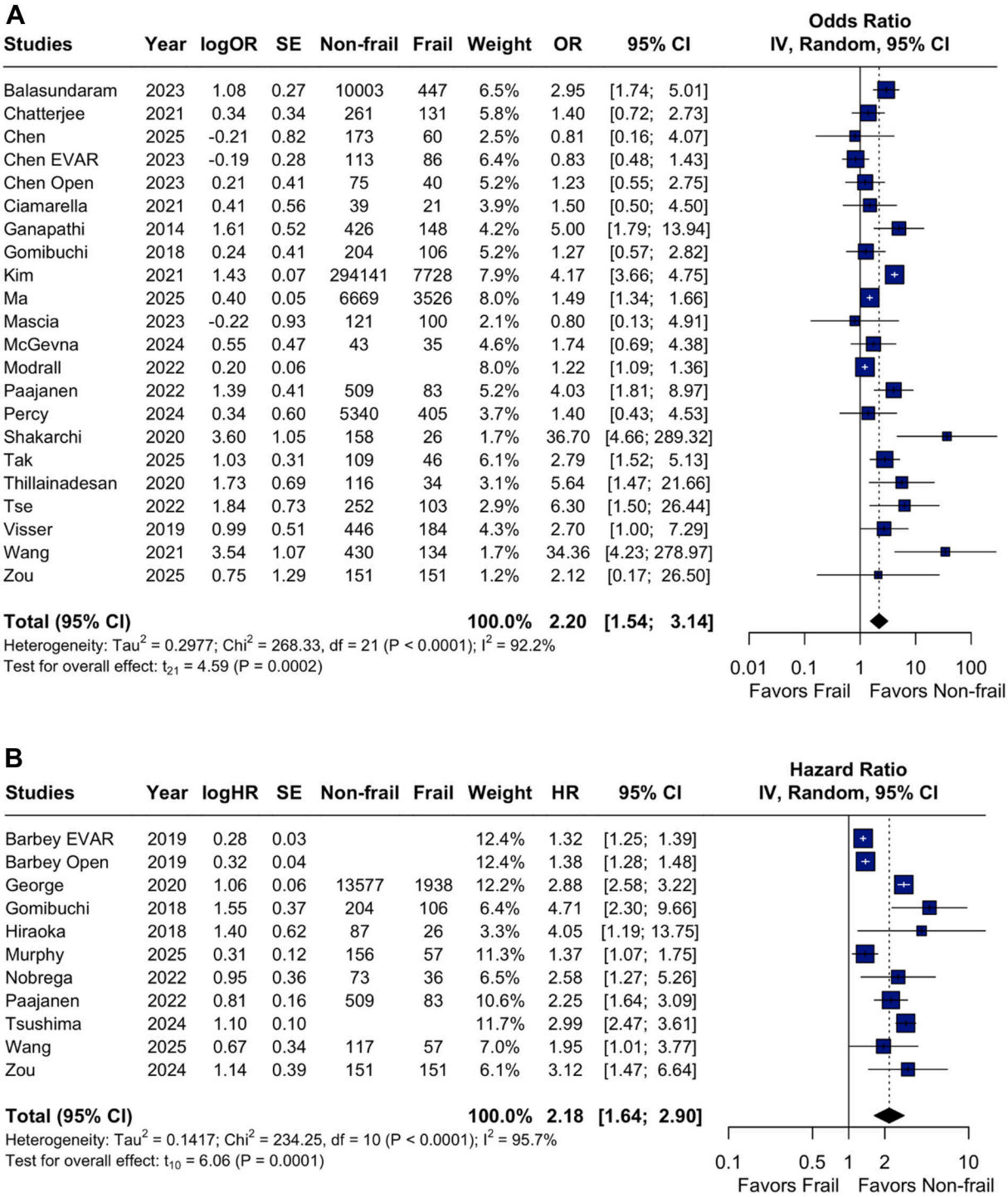


Fig. 2. Frailty and mortality after aortic surgery. Forest plots for the association between frailty and **(A)**, early mortality and **(B)**, late mortality.

HR of 2.18 (95% CI 1.64–2.90; $P = 0.0001$; $I^2 = 95.7\%$; Fig. 2).

Regarding postoperative morbidity and clinical outcomes, frail patients experienced a significantly increased odds of major complications (OR 2.52; 95% CI 1.22–5.19; $P = 0.0207$; $I^2 = 99.6\%$; Supplementary Fig. S1) and AKI (OR 1.64; 95% CI 1.34–2.02; $P = 0.0003$; $I^2 = 11.9\%$;

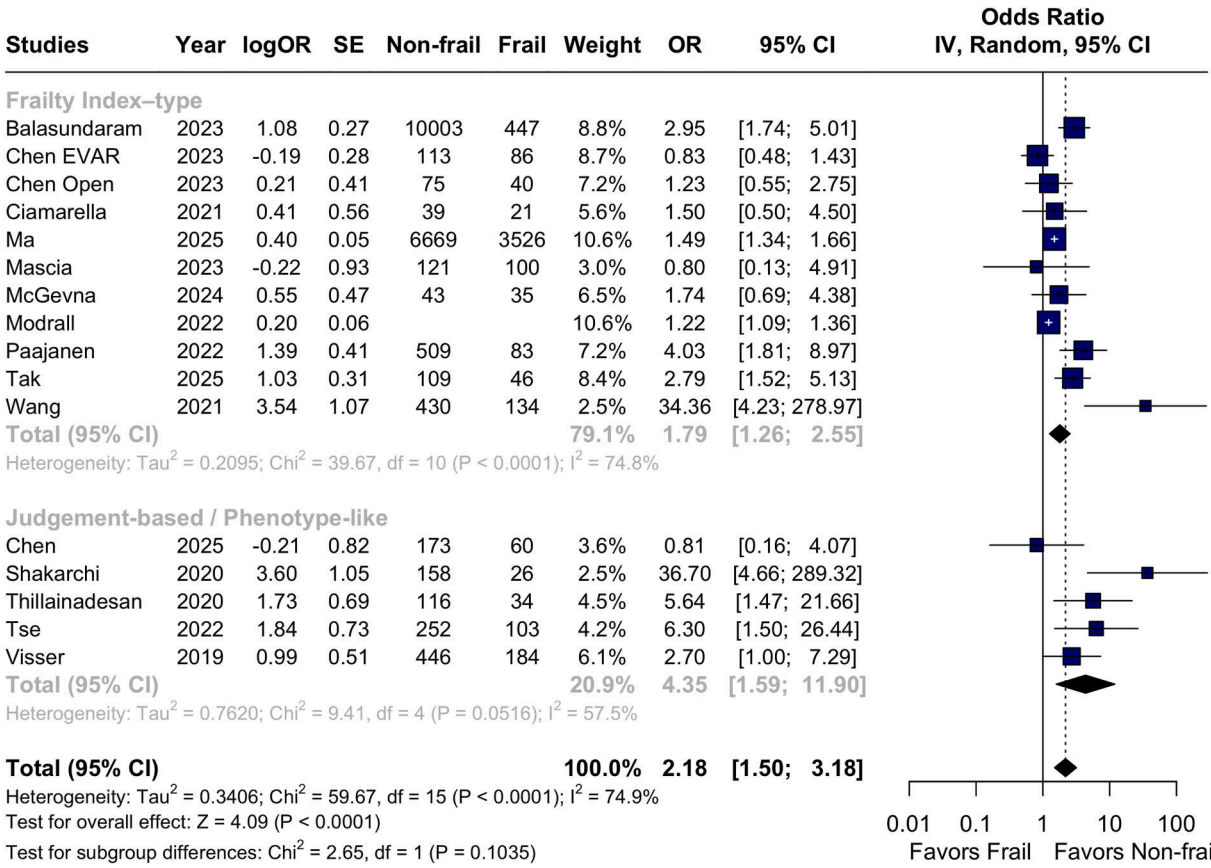


Fig. 3. Early mortality by frailty assessment tool. Subgroup analysis stratified by FI-type versus judgment-based/phenotype-like tools. FI, frailty index.

Supplementary Fig. S2). Furthermore, frailty was a strong predictor of nonhome discharge, with these patients having more than 5 times the odds of being discharged to a care facility or rehabilitation center rather than their primary residence (OR 5.50; 95% CI 3.05–9.92; $P = 0.0004$; $I^2 = 75.7\%$; Supplementary Fig. S3).

Subgroup Analyses

Subgroup analysis revealed that the association between frailty and early mortality was not influenced by the type of assessment tool used ($P = 0.1035$; Fig. 3). While both methodologies were significant predictors, judgment-based/phenotype-like assessments demonstrated higher odds of magnitude (OR 4.35; 95% CI 1.59–11.90; Fig. 3) compared to FI-type tools (OR 1.79; 95% CI 1.26–2.55; Fig. 3). However, subgroup differences within the FI-type category were not meaningful, as a secondary analysis showed no statistically significant difference between the mFI-11 and mFI-5 frameworks ($P = 0.1014$; Fig. 4). The mFI-11 was a robust predictor of early mortality (OR 2.39; 95% CI 1.46–3.92; Fig. 4), whereas the simplified mFI-5 did not reach statistical significance (OR 1.34; 95% CI 0.82–2.18; Fig. 4).

A subgroup analysis by aortic segment also demonstrated significant differences in the impact of frailty on early mortality ($P = 0.0074$; Supplementary Fig. S4). Frailty was a strong and significant predictor of mortality in patients undergoing AAA repair (OR 2.35; 95% CI 1.53–3.62; Supplementary Fig. S4) and TAAA repair (OR 1.22; 95% CI 1.10–1.36; Supplementary Fig. S4). Conversely, the association did not reach statistical significance for isolated thoracic procedures (OR 2.74; 95% CI 0.79–9.51; Supplementary Fig. S4), likely due to the limited number of studies and the resulting wide confidence intervals in this subgroup. This disparity was further confirmed by a focused analysis of the AAA cohort ($P = 0.0149$; Supplementary Fig. S5), where judgment-based/phenotype-like tools identified substantially higher

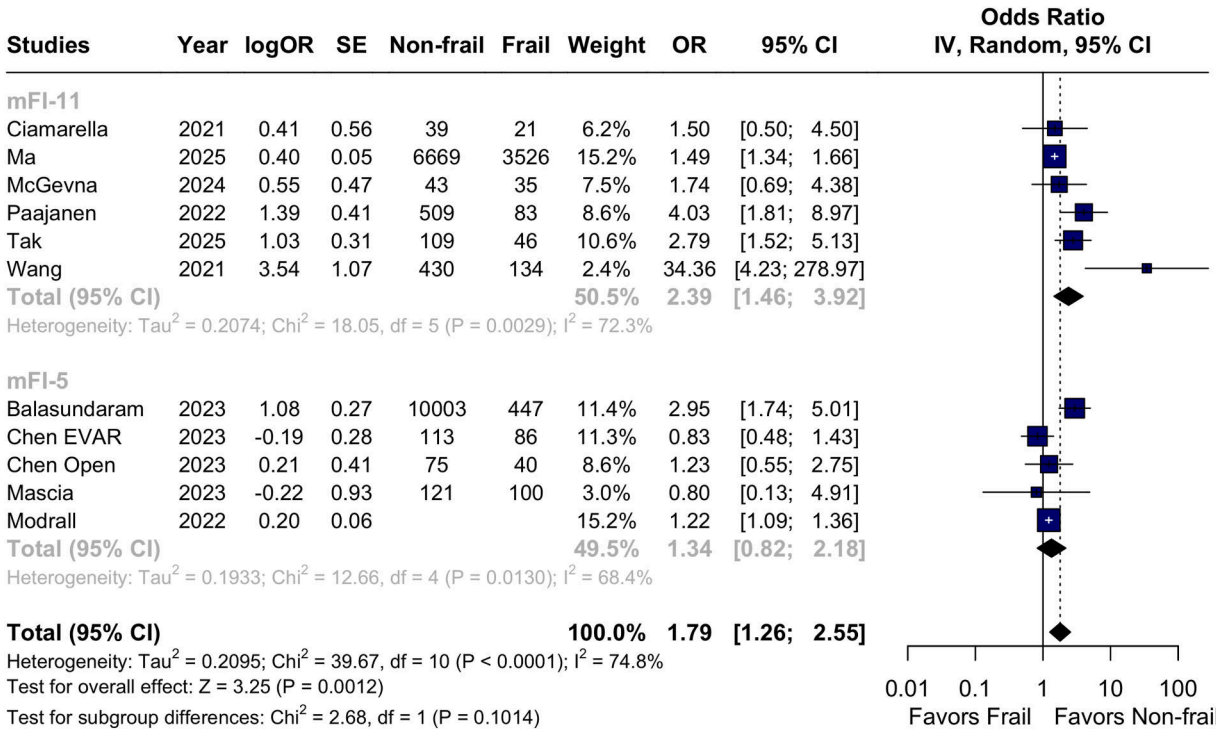


Fig. 4. Early mortality by Modified Frailty Index type. Subgroup analysis comparing mFI-5 and mFI-11. mFI, modified frailty index.

odds (OR 5.79; 95% CI 2.42−13.86; Supplementary Fig. S5) compared to FI-type tools (OR 1.76; 95% CI 1.19−2.61; Supplementary Fig. S5).

Finally, the impact of frailty on early mortality remained consistent regardless of the surgical approach ($P = 0.4833$; Supplementary Fig. S6). Frailty was significantly associated with increased mortality in both endovascular (OR 2.63; 95% CI 1.4−4.72; Supplementary Fig. S6) and open surgical repairs (OR 1.94; 95% CI 1.05−3.60; Supplementary Fig. S6), suggesting that the predictive value of frailty is independent of the surgical technique employed.

Sensitivity Analyses

In the leave-one-out analyses for major complications, exclusion of the studies by Tse et al.[47] and Kim et al.[33] changed the results from significant to nonsignificant, without meaningfully affecting heterogeneity. For the remaining evaluated outcomes, removal of any individual study did not influence either the heterogeneity or the results. Detailed leave-one-out analyses are detailed in Supplementary Figures S7−S10.

In the analysis excluding studies with serious risk of bias, frailty remained associated with significantly higher odds of short-term mortality following aortic surgery. The pooled analysis yielded an OR of 1.43 (95% CI 1.04−1.97; $P = 0.0356$; $I^2 = 61.7\%$; Supplementary Fig. S11), with reduced heterogeneity compared with the overall analysis, without changing the direction or significance of the results.

Risk of Bias and Quality Assessment

Overall, the risk of bias across the included studies was predominantly moderate, with only a few studies classified as having a serious overall risk of bias. Most studies demonstrated low risk in the domains related to classification of interventions (D2) and measurement of outcomes (D5). In contrast, concerns were more frequent in the domains addressing confounding (D1), selection of participants (D3), missing data (D4), and selective reporting (D6), which were commonly rated as moderate risk. Serious risk ratings were observed in a limited number of studies, mainly due to issues related to participant selection and missing data. Overall, the methodological quality of the included

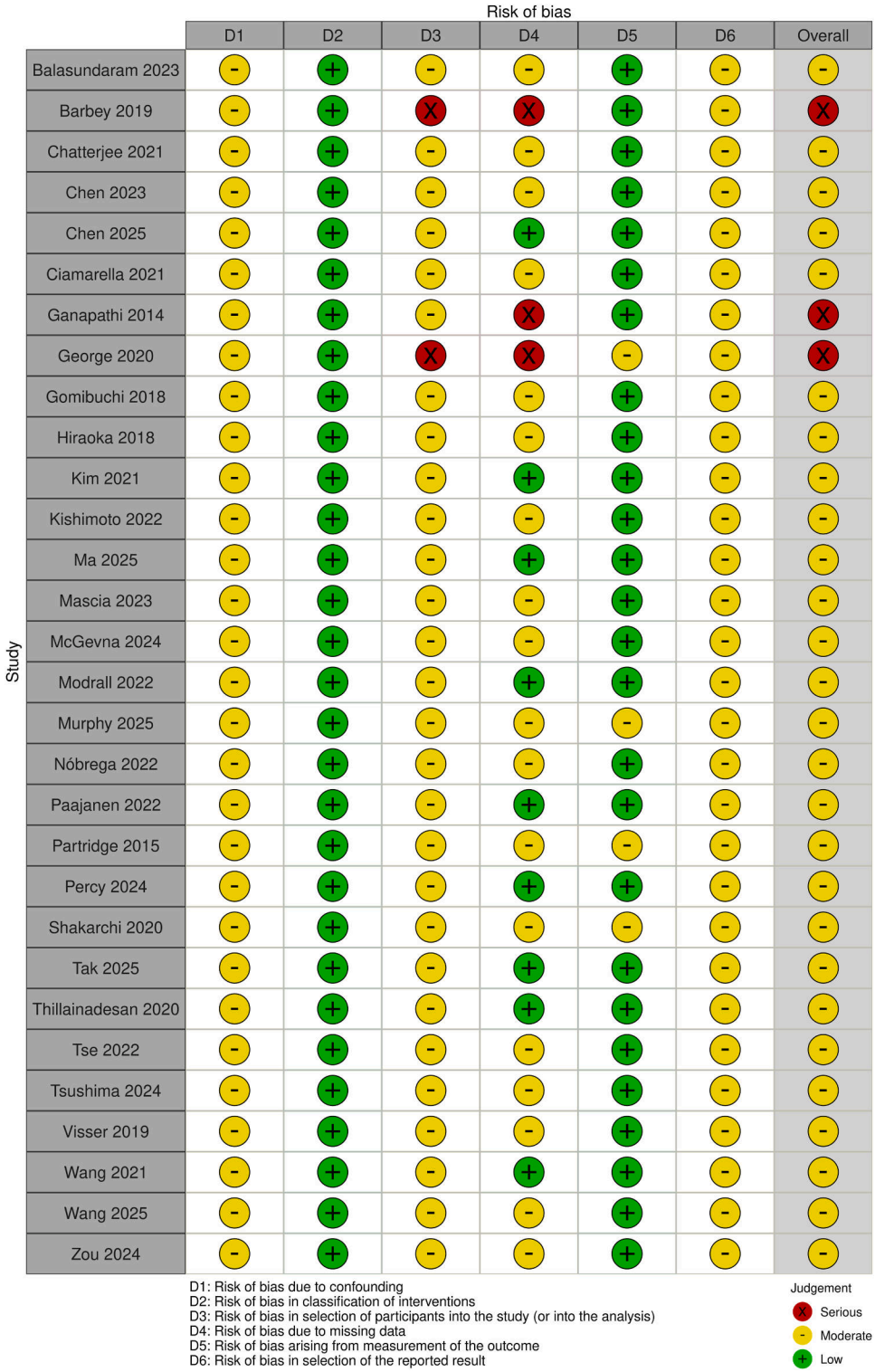


Fig. 5. Risk of bias assessment. ROBINS-I assessment across included studies. ROBINS-I, risk of bias in nonrandomized studies of interventions.

studies was acceptable, although some limitations reduce the certainty of the evidence. The detailed risk of bias assessment is presented in Figure 5.

The GRADE certainty of evidence was very low across all outcomes due to study design limitations, and indirectness, as detailed in Supplementary

Table II. All included studies were retrospective and nonrandomized, with moderate to serious confounding risk. Additional downgrading was applied for inconsistency (substantial heterogeneity, I^2>75%). For publication bias assessment of early mortality, visual inspection of the funnel plot revealed no clear asymmetry (Supplementary Fig. S12). This was supported by Egger's regression test, which showed no evidence of small study effects or funnel plot asymmetry (t = 0.73, df = 20, P = 0.4766; bias estimate = 0.7063, Standard error = 0.9737).

DISCUSSION

Our meta-analysis, including 30 studies and 419,459 patients, reinforces the robust association between frailty and adverse outcomes after aortic surgery, even when accounting for differences in study populations, operative techniques, and frailty definitions. Frail patients in our analysis had roughly twice the odds of dying in the early postoperative period and during long-term follow-up (OR 2.20 and HR 2.18, respectively). This finding is consonant with broader surgical evidence; a review noted that a meta-analysis of 16,679 cardiac surgical patients reported a 2- to 5-fold increase in postoperative complications, mortality, and readmissions among frail patients,[7] and a meta-analysis of 1.1 million adults showing that frailty raises postoperative complications by 2.4-fold and mortality by 3.7-fold.[53] These parallels suggest that frailty is not merely a marker of immediate perioperative risk but reflects systemic vulnerability that influences survival long after discharge.

Our data also highlight how the chosen frailty tool affects risk estimates. Phenotype- or judgment-based instruments yielded larger ORs than deficit accumulation indices, although the difference was not statistically significant. This echoes findings from cardiac surgery, where assessments such as gait speed or the CFS are strong mortality predictors.[54] Within deficit-based tools, the mFI-11 captured higher odds than the abbreviated mFI-5, supporting the notion that more comprehensive indices offer greater discriminatory power. Lack of standardization remains a central limitation; the 2024 AAA meta-analysis found that frailty prevalence ranged from 2.3% to 34.6%, and that frail patients had a 3.20-fold higher risk of short-term death and a 2.86-fold higher risk of long-term mortality.[5] Harmonizing frailty definitions would improve comparability across studies.

Clinical context modifies the effect of frailty. Stratification by aortic segment showed that the impact was strongest for AAA (OR 2.35) and more modest for thoracoabdominal aneurysm (OR 1.22). The diminished effect in thoracoabdominal cases may reflect that these patients already face very high operative risk, leaving less room for frailty to discriminate. Procedural approach did not significantly modify the association (OR 2.63 for endovascular versus 1.94 for open repair), suggesting that frailty conveys prognostic information independent of surgical technique. Objective complications were differentially affected; overall complications showed an unstable pooled estimate with extreme heterogeneity, whereas AKI was consistently 64% more common in frail patients with very low heterogeneity, indicating that frailty reliably predicts organ-specific dysfunction when outcomes are well defined. These patterns imply that heterogeneity in composite outcomes arises more from inconsistent definitions than from biological variability.

Functional outcomes are particularly sensitive to frailty. Our analysis found that frail patients were more than 5 times as likely to require discharge to a facility rather than home. Systematic reviews of vascular surgery report similar findings; frail patients have roughly 2.5-fold higher risk of nonhome discharge underscoring that loss of independence may be the most profound consequence of frailty.[5] This has important implications for shared decision-making; patients often prioritize functional recovery over survival, and a realistic discussion of discharge destination should accompany risk counseling. The biological plausibility of these findings is clear; frailty reflects diminished reserve across multiple systems, leading to greater susceptibility to organ dysfunction, slower recovery, and prolonged institutionalization.

Emerging evidence suggests that targeted optimization may mitigate some of these risks. A 2025 meta-analysis of exercise-based prehabilitation in cardiac surgery found that prehabilitation significantly improved functional capacity, reduced respiratory complications, and shortened hospital stay.[55] Prehabilitation programs lasting at least 4 weeks conferred the greatest benefits, supporting the concept that enhancing physiological reserve before surgery can improve outcomes. The 2024 American Heart Association/American College of Cardiology perioperative guideline now gives a class IIa recommendation for frailty assessment and notes that nutritional and exercise prehabilitation may reduce pulmonary and cardiac complications. Integrating frailty screening with multimodal optimization, including nutrition, exercise, and geriatric comanagement, could therefore translate predictive insight into improved perioperative care.[4]

Sarcopenia may further refine this risk stratification process as an objective, imaging-based surrogate of frailty. Mezzetto et al. showed that computed tomography angiography–based skeletal muscle assessment, most commonly using psoas muscle area at L3 or L4, can identify sarcopenia among patients undergoing endovascular aneurysm repair (EVAR), thoracic endovascular aortic repair, or complex EVAR. Despite heterogeneity in definitions and thresholds, sarcopenia was highly prevalent and was associated with higher mortality, adverse events, and poorer long-term survival, suggesting that routinely available preoperative imaging may complement clinical frailty assessment in aortic surgery.[56]

In parallel, the 2026 European Society for Vascular Surgery guidelines on descending thoracic and thoracoabdominal aortic diseases emphasize multidisciplinary risk assessment, medical optimization, shared decision-making, and treatment individualization based on life expectancy and functional status. In complex aortic repair, where chronological age alone does not fully capture operative vulnerability, integrating frailty screening with nutritional, exercise-based, cardiovascular, and geriatric optimization may support individualized perioperative planning and improve postoperative recovery.[11]

Several limitations temper these conclusions. Heterogeneity was high for several outcomes, reflecting differences in study design, patient populations, frailty definitions, and adjustment for confounders. Many included studies were retrospective and varied in whether outcomes were adjusted for age or comorbidities, introducing potential bias. Important confounders such as urgent versus elective surgery, extent of aortic disease, and postoperative care protocols were not uniformly reported. Furthermore, the predominance of deficit-based tools like the mFI may underestimate or overestimate risk depending on the patient population, and phenotype-based tools were less common, limiting comparisons. Finally, data were insufficient to evaluate interventions such as prehabilitation in the context of aortic surgery; most trials are in cardiac or cancer populations and may not directly translate to complex vascular procedures.

CONCLUSION

Despite some limitations, the weight of evidence seems to support frailty as a potentially clinically meaningful predictor of mortality, organ-specific complications and, importantly, postoperative independence. Routine preoperative frailty assessment can identify patients at high risk of adverse and should prompt a multidisciplinary discussion about surgical candidacy, potential need for rehabilitation services, and optimization strategies. Future research should focus on standardizing frailty definitions, incorporating frailty into risk calculators, exploring the efficacy of prehabilitation in aortic surgery, and conducting meta-regressions adjusting for baseline risk factors. By moving from risk recognition to risk modification, clinicians can better align treatment with patient goals and improve outcomes in this vulnerable population.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Mayara Leite Coutinho: Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lucas Monteiro Delgado:** Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Inez Ohashi Torres:** Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **João Rocha-Neves:** Writing — review & editing, Writing — original draft, Supervision, Software, Investigation. **Antonio Eduardo Zerati:** Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mario D'Oria:** Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pedro Puech Leão:** Writing — review & editing, Writing — original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found at https://doi.org/10.1016/j.avsg.2026.06.034.

REFERENCES

1. Houghton JSM, Nickinson ATO, Morton AJ, et al. Frailty factors and outcomes in vascular surgery patients: a

systematic review and meta-analysis. Ann Surg Agosto de 2020;272:266–76.
2. Wang J, Zou Y, Zhao J, et al. The impact of frailty on outcomes of elderly patients after major vascular surgery: a systematic review and meta-analysis. Eur J Vasc Endovasc Surg 2018;56:591–602.
3. Sieber F, McIsaac DI, Deiner S, et al. 2025 American Society of Anesthesiologists Practice Advisory for perioperative care of older adults scheduled for inpatient surgery. Anesthesiology 2025;142:22–51.
4. Thompson A, Fleischmann KE, Smilowitz NR, et al. 2024 AHA/ACC/ACS/ASNC/HRS/SCA/SCCT/SCMR/SVM Guideline for perioperative cardiovascular management for noncardiac surgery: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 2024;150: e351–442.
5. Wang YX, Zhao WX, Wang BR, et al. The effect of frailty on prognosis of patients undergoing abdominal aortic aneurysm repair: a systematic review and meta-analysis. Ann Vasc Surg 2026;122:476–85.
6. Xu F, Mo J, Chen X. Role of frailty in predicting outcomes after surgical repair of abdominal aortic aneurysm: a systematic review and meta-analysis. Ann Vasc Surg 2025;120: 223–34.
7. Wong CWY, Yu DSF, Li PWC, et al. The prognostic impacts of frailty on clinical and patient-reported outcomes in patients undergoing coronary artery or valvular surgeries/procedures: a systematic review and meta-analysis. Ageing Res Rev 2023;85:101850.
8. Aucoin SD, Hao M, Sohi R, et al. Accuracy and feasibility of clinically applied frailty instruments before surgery: a systematic review and meta-analysis. Anesthesiology 2020;133:78–95.
9. Ijaz N, Buta B, Xue QL, et al. Interventions for frailty among older adults with cardiovascular disease: JACC state-of-the-art review. J Am Coll Cardiol 2022;79:482–503.
10. Wanhainen A, Van Herzeele I, Bastos Goncalves F, et al. Editor's choice – European Society for Vascular Surgery (ESVS) 2024 Clinical Practice Guidelines on the management of abdominal aorto-iliac artery aneurysms. Eur J Vasc Endovasc Surg 2024;67:192–331.
11. Wanhainen A, Gombert A, Antoniou GA, et al. European Society for Vascular Surgery (ESVS) 2026 Clinical Practice Guidelines on the management of descending thoracic and thoraco-abdominal aortic diseases - editor's choice. Eur J Vasc Endovasc Surg 2026;71:172–270.
12. Thangavelu V, Bhatia O, Hasija A, et al. Association of preoperative frailty and postoperative outcomes in older adults undergoing major cardiac procedures: a systematic review and meta-analysis. Anesth Analg 2025;. https://doi.org/10.1213/ANE.0000000000007887.
13. Kim DH, Kim CA, Placide S, et al. Preoperative frailty assessment and outcomes at 6 months or later in older adults undergoing cardiac surgical procedures: a systematic review. Ann Intern Med 2016;165:650–60.
14. McGinigle KL, Spangler EL, Pichel AC, et al. Perioperative care in open aortic vascular surgery: a consensus statement by the Enhanced Recovery After Surgery (ERAS) Society and Society for Vascular Surgery. J Vasc Surg 2022;75: 1796–820.
15. Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for Systematic Reviews of Interventions version 6.4. Available at: www.training.cochrane.org/handbook; 2023. Accessed April 20, 2024.
16. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71.
17. U.S. National Institute for Health and Care Research. PROSPERO: International Prospective Register of Systematic Reviews. Available at: https://www.crd.york.ac.uk/prospero/. Accessed April 20, 2024.
18. Wohlin C. Guidelines for snowballing in systematic literature studies and a replication in software engineering [Internet]. Available at: https://dl.acm.org/doi/10.1145/2601248.2601268. Accessed April 20, 2024.
19. Viechtbauer W, Cheung MWL. Outlier and influence diagnostics for meta-analysis. Res Synth Methods 2010;1: 112–25.
20. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ 2016;355:i4919.
21. Atkins D, Best D, Briss PA, et al. Grading quality of evidence and strength of recommendations. BMJ 2004;328:1490.
22. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. BMJ 1997;315: 629–34.
23. Balasundaram N, Chandra I, Sunilkumar VT, et al. Frailty Index (mFI-5) predicts resource utilization after nonruptured endovascular aneurysm repair. J Surg Res 2023;283: 507–13.
24. Barbey SM, Scali ST, Kubilis P, et al. Interaction between frailty and sex on mortality after elective abdominal aortic aneurysm repair. J Vasc Surg 2019;70:1831–43.
25. Chatterjee S, Shi A, Yoon L, et al. Effect of sarcopenia on survival and spinal cord deficit outcomes after thoracoabdominal aortic aneurysm repair in patients 60 years of age and older. J Thorac Cardiovasc Surg 2023;165:1985–1996.e3.
26. Chen AJ, Yeh SL, Ulloa JG, et al. Frailty among veterans undergoing abdominal aortic aneurysm repair. Ann Vasc Surg 2023;92:18–23.
27. Chen S, Ramirez E, Murphy BE, et al. Increased clinical frailty is associated with aortic-related mortality following fenestrated and branched endovascular repair for thoracoabdominal aortic aneurysm. J Vasc Surg 2026;83:688–98.
28. Ciaramella M, Kravchenko T, Grieff A, et al. Frailty is a poor predictor of postoperative morbidity and mortality after ruptured abdominal aortic aneurysm. Ann Vasc Surg 2021;74:122–30.
29. Ganapathi AM, Englum BR, Hanna JM, et al. Frailty and risk in proximal aortic surgery. J Thorac Cardiovasc Surg 2014;147:186–191.e1.
30. George EL, Chen R, Trickey AW, et al. Variation in center-level frailty burden and the impact of frailty on long-term survival in patients undergoing elective repair for abdominal aortic aneurysms. J Vasc Surg 2020;71:46–55.e4.
31. Gomibuchi T, Seto T, Komatsu M, et al. Impact of frailty on outcomes in acute type A aortic dissection. Ann Thorac Surg 2018;106:1349–55.
32. Hiraoka A, Saito K, Chikazawa G, et al. Modified predictive score based on frailty for mid-term outcomes in open total aortic arch surgery. Eur J Cardiothorac Surg 2018;54:42–7.
33. Kim ST, Tran Z, Hadaya J, et al. Impact of frailty on acute outcomes of endovascular thoracic and abdominal aneurysm repair. Surgery 2021;170:304–10.
34. Kishimoto Y, Yoshikawa Y, Morimoto K, et al. Impact of frailty on early and mid-term outcomes of hybrid aortic arch repair. Surg Today 2022;52:1194–201.
35. Ma Y, Gittinger M, Nguyen T, et al. A multivariate analysis of a modified frailty index on perioperative morbidity and

mortality following nonemergent endovascular aortic aneurysm repair. Ann Vasc Surg 2025;113:138–47.
36. Mascia D, Santoro A, Saracino C, et al. Five-factors modified frailty index role as predictors of outcomes after proximal abdominal aortic aneurysms. Int Angiol J Int Union Angiol 2023;42:520–7.
37. McGevna MA, Adler LSF, Lu JY, et al. The modified frailty index-11 is a poor predictor of 1-year mortality and morbidity after ruptured abdominal aortic aneurysm repair. Ann Vasc Surg 2025;120:413–21.
38. Modrall JG, Tsai S, Ramanan B, et al. Frailty as a predictor of mortality for fenestrated EVAR and open surgical repair of aortic aneurysms involving visceral vessels. Ann Vasc Surg 2022;80:29–36.
39. Murphy BE, Bunker M, Gillan A, et al. Clinical frailty predicts long-term survival and return to functional status following fenestrated and branched aortic repair for thoracoabdominal aortic aneurysm. J Vasc Surg 2025;82:62–71.
40. Nóbrega L, Pereira-Neves A, Duarte-Gamas L, et al. Outcome analysis using the modified frailty index-5 in patients with complex aortoiliac disease. Ann Vasc Surg 2022;79:153–61.
41. Paajanen P, Kärkkäinen JM, Tenorio ER, et al. Effect of patient frailty status on outcomes of fenestrated-branched endovascular aortic repair for complex abdominal and thoracoabdominal aortic aneurysms. J Vasc Surg 2022;76:1170–1179.e2.
42. Partridge JSL, Fuller M, Harari D, et al. Frailty and poor functional status are common in arterial vascular surgical patients and affect postoperative outcomes. Int J Surg 2015;18:57–63.
43. Percy ED, Faggion Vinholo T, Newell P, et al. The impact of frailty on outcomes of Proximal aortic aneurysm surgery: a nationwide analysis. J Cardiovasc Dev Dis 2024;11:32.
44. Al Shakarchi J, Fairhead J, Rajagopalan S, et al. Impact of frailty on outcomes in patients undergoing open abdominal aortic aneurysm repair. Ann Vasc Surg 2020;67:100–4.
45. Tak S, Yaman O, lrz E, et al. The role of frailty index in predicting postoperative complications IN geriatric vascular surgery patients. Turk J Geriatr 2025;28:482–92.
46. Thillainadesan J, Mudge AM, Aitken SJ, et al. The prognostic performance of frailty for delirium and functional decline in vascular surgery patients. J Am Geriatr Soc 2021;69:688–95.
47. Tse W, Lavingia KS, Amendola MF. Using the risk analysis index to assess frailty in a veteran cohort undergoing endovascular aortic aneurysm repair. J Vasc Surg 2022;75:1591–1597.e1.
48. Tsushima S, Shibata T, Iba Y, et al. Long-Term outcomes and late complications of thoracic endovascular aortic repair using the GORE TAG or conformable GORE TAG. J Endovasc Ther 2024;33:15266028241305960.
49. Visser L, Banning LBD, El Moumni M, et al. The effect of frailty on outcome after vascular surgery. Eur J Vasc Endovasc Surg 2019;58:762–9.
50. Wang J, Zhao J, Ma Y, et al. Frailty as a predictor of major adverse cardiac and cerebrovascular events after endovascular aortic aneurysm repair. J Vasc Surg 2021;74:442–450.e4.
51. Wang YX, Zhao WX, Wang ZM, et al. Frailty impacts all-cause mortality after endovascular abdominal aortic aneurysm repair: a retrospective cohort study. J Nutr Health Aging 2025;29:100489.
52. Zou Y, Wang J, Zhao J, et al. Predictive value of geriatric nutritional risk index in cardiac and cerebrovascular events after endovascular aortic aneurysm repair. Front Cardiovasc Med 2024;11:1399908.
53. Tjeertes EKM, van Fessem JMK, Mattace-Raso FUS, et al. Influence of frailty on outcome in older patients undergoing non-cardiac surgery - a systematic review and meta-analysis. Aging Dis 2020;11:1276–90.
54. Chang J, Nathalie J, Nguyenhuy M, et al. Slow gait speed is associated with worse postoperative outcomes in cardiac surgery: a systematic review and meta-analysis. J Card Surg 2022;37:197–204.
55. Hurtado-Borrego JC, Bayonas-Ruiz A, Bonacasa B. Exercise-Based prehabilitation before cardiac surgery: a systematic review, meta-analysis, meta-regression, and proposal for a clinical implementation model. J Clin Med 2025;14:8195.
56. Mezzetto L, D'Oria M, Mani K, et al. Scoping review of radiologic assessment and prognostic impact of skeletal muscle sarcopenia in patients undergoing endovascular repair for aortic disease. J Vasc Surg 2022;76:1407–16.